

# Comparative study between a GA-tuned PID controller based on minimizing IAE, ITAE, and ISE objective functions in a shell and tube heat exchanger temperature control system

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## Abstract

A heat exchanger device is widely used in process industries because it is capable of sustaining a wide range of temperatures. The main purpose of a heat exchanger is to transfer heat from a hot fluid to a cooler fluid so that the temperature of the process fluid is controlled. Among the different heat exchanger types, shell and tube heat exchangers are the most commonly used type in the process industry. The heat exchanger temperature control system is a highly nonlinear, uncertain, time-delayed, and complex system. In addition to that, it is accompanied by the presence of two predominant disturbances, namely flow variation and temperature variation of input fluid. In this study, the performance of a genetic algorithm-based proportional integral and derivative controller tuned based on minimizing IAE, ITAE, and ISE objective functions has been analyzed and compared under four scenarios, namely no disturbance, flow variation disturbance, temperature variation disturbance, and both disturbance conditions. Among the three objective functions used, the overall performance of the genetic algorithm-based proportional integral and derivative controller tuned based on minimizing the time integral of the absolute error objective function is better than the one tuned based on minimizing the integral of the absolute error and the integral of the squared error objective functions.

## Keywords

ga tuned pid controller, genetic algorithm, integral square error, power, energy and industry applications

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## **ABSTRACT**

*A heat exchanger device is widely used in process industries because it is capable of sustaining a wide range of temperatures. The main purpose of a heat exchanger is to transfer heat from a hot fluid to a cooler fluid so that the temperature of the process fluid is controlled. Among the different heat exchanger types, shell and tube heat exchangers are the most commonly used type in the process industry. The heat exchanger temperature control system is a highly nonlinear, uncertain, time-delayed, and complex system. In addition to that, it is accompanied by the presence of two predominant disturbances, namely flow variation and temperature variation of input fluid. In this study, the performance of a genetic algorithm-based proportional integral and derivative controller tuned based on minimizing IAE, ITAE, and ISE objective functions has been analyzed and compared under four scenarios, namely no disturbance, flow variation disturbance, temperature variation disturbance, and both disturbance conditions. Among the three objective functions used, the overall performance of the genetic algorithm-based proportional integral and derivative controller tuned based on minimizing the time integral of the absolute error objective function is better than the one tuned based on minimizing the integral of the absolute error and the integral of the squared error objective functions.*

**Key words:** Genetic algorithm, GA tuned PID controller, Integral Square Error, Integral Absolute Error and Integral Time Absolute Error.

# 1. Introduction

A heat exchanger (HE) is a thermal device used in process industries as a medium to transfer heat energy (enthalpy) from hot fluid to cooler fluid so that the targeted temperature of the process fluid is attained [1]. The mechanism through which the transfer of energy takes place is indirect; there is no direct physical contact between the two fluids [2]. These devices are crucial in the process industry, which has attracted the attention of many researchers in studying HE system control [3]. In the STHE system, the manipulated (controlling) fluid flows through the tube side and the controlled fluid flows through the shell side of the STHE [4].

The heat exchanger system is a highly nonlinear, uncertain, time-delayed, and complex system. In addition to that, it is accompanied by the presence of two predominant disturbances, namely: flow variation of the input fluid and temperature variation of the input fluid [5]. Achieving desired system performance despite these challenges, forced control engineers to design an advanced control strategy [6].

HE is a critical device in the process industry; it is used to preheat and pasteurize the process variable. Such activities are achieved within certain ranges of temperature and affect the flavor, shelf life, nutritional content, and quality of the product. Even though the device is related to safety, industries are still using a classical PID controller to control the temperature of the exchanger device [7]. In this study, the parameters of proportional, integral, and derivative gains are tuned using three different objective functions, namely IAE, ITAE, and ISE, so that industries can choose based on the performance they need to achieve.

## 1.1 Heat transfer mechanism of heat exchanger system

The heat exchanger uses conduction and convection so that heat energy is transferred between two fluids. This is achieved based on the principle that heat energy is transferred by convection from the hotter fluid to the separating medium, which separates the two fluids. Then the separating medium conducts energy to the other side of it by conduction. Finally, the cooler fluid takes away heat energy from the separating wall by convection [8]. The driving force for heat energy transfer in an HE device is temperature deviation. This temperature deviation is expressed in terms of a log mean temperature difference (LMTD). The greater the LMTD, the more heat will be transferred. For a better understanding of LMTD, let us take a look at figure 1.1[9].

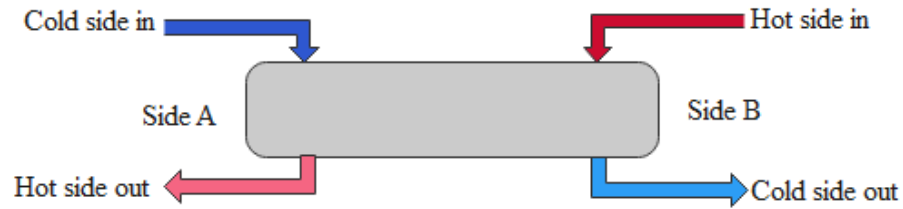


Figure 1. 1 Log mean temperature difference

The LMTD is represented by the equation below.

$$LMTD = \frac{\Delta T_A - \Delta T_B}{\log\left(\frac{\Delta T_A}{\Delta T_B}\right)} \quad (1.1)$$

Where,  $\Delta T_A$  is a difference in temperature for side A, and  $\Delta T_B$  is a difference in temperature for side B.

## 2. Schematic and block diagram of the proposed system

### 2.1 Schematic diagram of the proposed system

As can be observed on the schematic diagram, two fluids are being supplied to the system. Superheated steam is supplied from the boiler to the tube side, and process fluid is supplied from the storage tank to the shell side of the STHE [10].

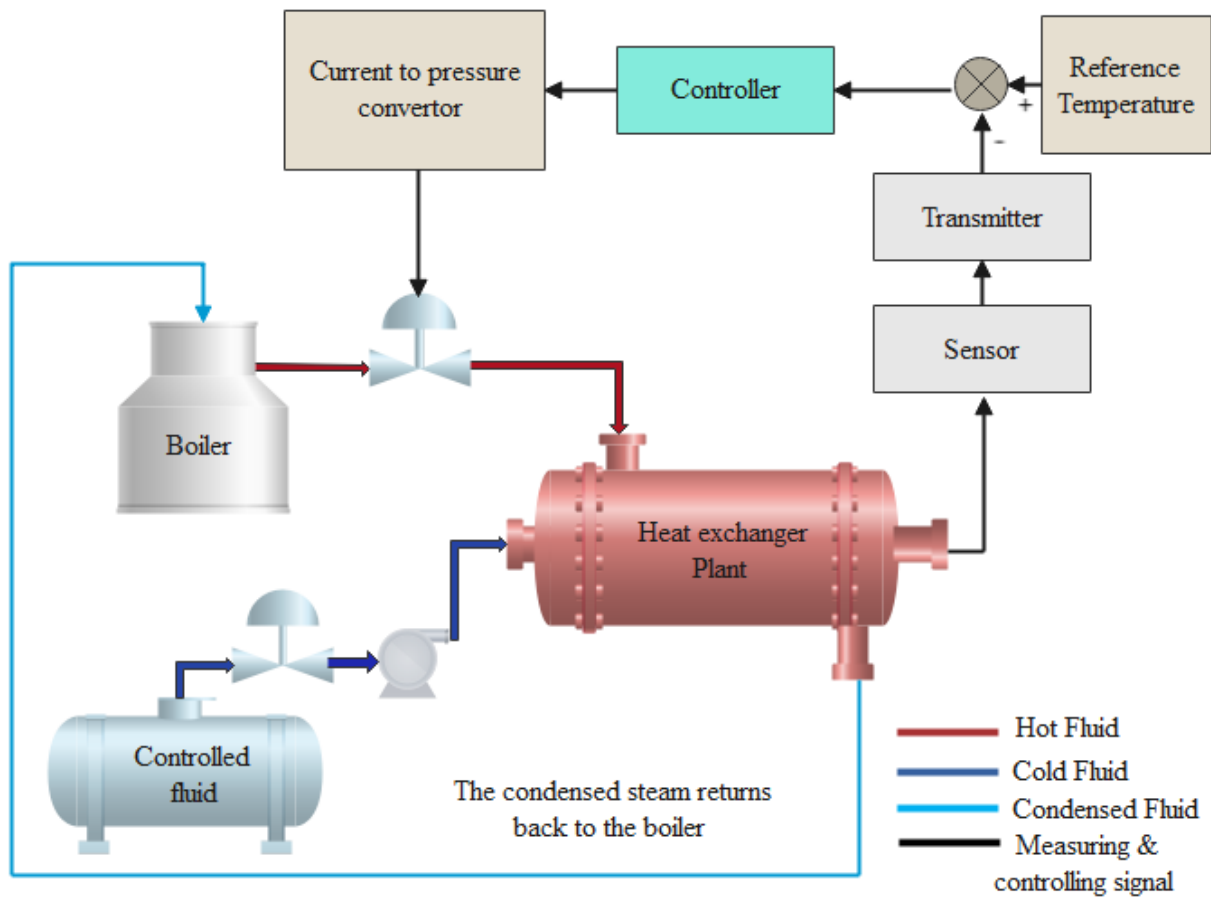


Figure 2. 1 Schematic diagram of the proposed system

Here, the main objective is to regulate and sustain the temperature of the process fluid at the desired set point (reference temperature) by varying the flow rate of the steam, and this is achieved based on the following principle: the outlet temperature of the process fluid that is measured by the thermocouple temperature sensor is compared with the desired temperature. The deviation between the two is calculated, which is going to be fed as an input to the proposed controller. Since the signal generated by the controller is a current signal, it is converted to a pressure signal via a current to pressure converter. Finally, this pressure signal positions the valve in proportion to the control signal.

## 2.2 block diagram of the proposed system

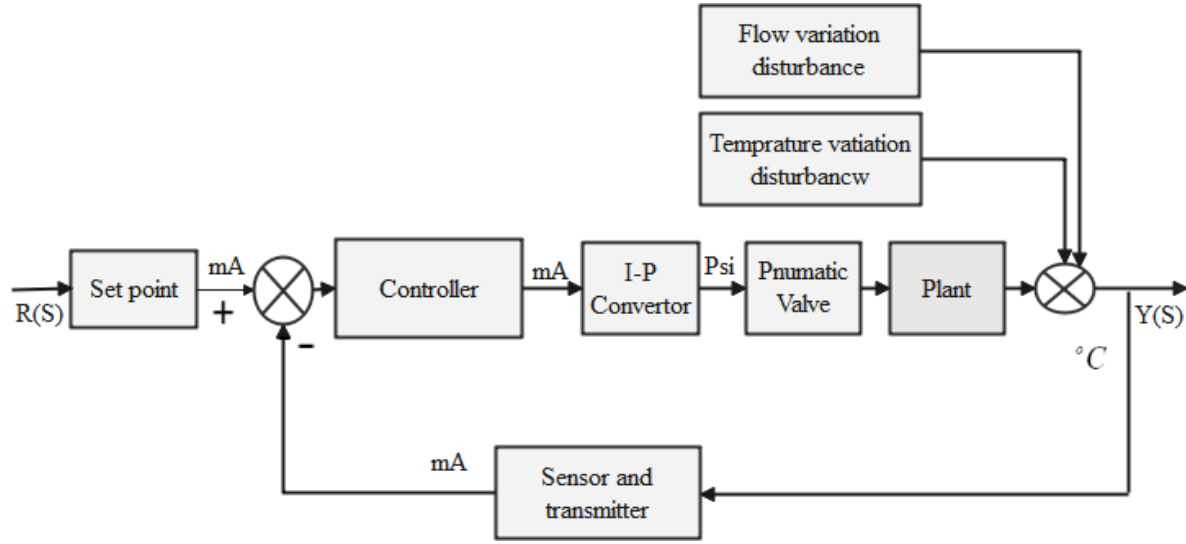


Figure 2. 2 Block diagram of the proposed system [10]

This study considers the experimental data mentioned in literature [11][10] while developing the transfer function model of heat exchanger system.

The transfer function of the plant:  $G_p(s) = \frac{50e^{-s}}{30s+1}$  (2.1)

Transfer function of valve:  $G_v = \frac{0.13}{3s+1}$  (2.2)

Transfer function of sensor:  $G_s(s) = \frac{0.16}{10s+1}$  (2.3)

Transfer function of flow variation disturbance:  $G_{dF}(s) = \frac{1}{30s+1}$  (2.4)

Transfer function of temperature variation disturbance:  $G_{dT}(s) = \frac{3}{30s+1}$  (2.5)

$I - P$  converter = 0.75 Psi/mA

The overall transfer function of the plant with the control valve is represented by SOPDT as shown in equation 2.6:

$$G(s) = \frac{4.875e^{-\tau_d s}}{90s^2 + 33s + 1} \quad (2.6)$$

### 3. Controller design

#### 3.1 PID controller

Around 90% of the control loops in the process industry are PID control loops [5]. This is due to its simplicity, cost-effectiveness, and ease of implementation [12]. The PID controller may have either a parallel or series arrangement. The parallel arrangement is shown in figure 3.1. In a PID controller, the proportional term generates a control signal which is proportional to the error signal boost the response, whereas the derivative term improves the transient response (stability) and the integral term eliminates steady-state error [13].

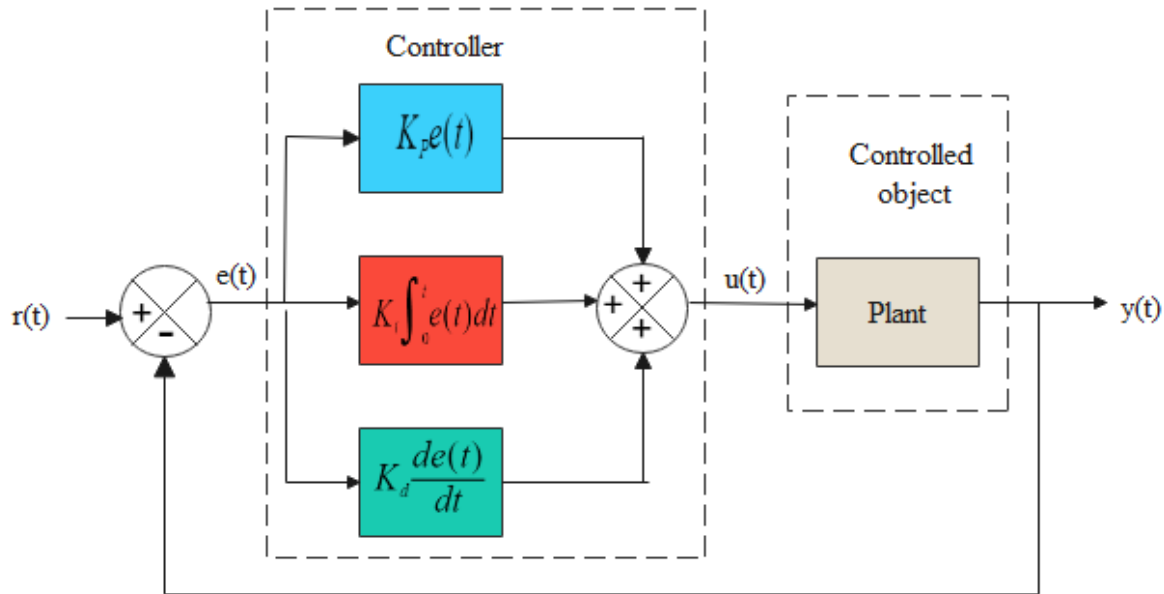


Figure 3. 1 Structure of PID controller [14]

Where  $K_p$  is the proportional gain,  $K_i$  is integral gain,  $K_d$  is derivative gain  $r(t)$  is set point,  $e(t)$  is the error signal, and  $y(t)$  is the output of the controlled system. The error signal  $e(t)$  is the deviation between the actual output and the desired output by the process (set point). The time domain representation for the parallel PID controller structure are shown in equation (3.1) [15][16].

$$U(t) = \underbrace{K_p e(t)}_{\text{Proportional action}} + \underbrace{K_i \int e(t) dt}_{\text{Integral action}} + \underbrace{K_d \frac{de(t)}{dt}}_{\text{Derivative action}} \quad (3.1)$$

Table 3. 1 Table Effect of PID gains on steady and transient state performance

| Gain         | Overshoot | Settling time | Steady state error |
|--------------|-----------|---------------|--------------------|
| Proportional | Increase  | Small change  | Decrease           |
| Integral     | Increase  | Increase      | Decrease           |
| Derivative   | Decrease  | Decrease      | No change          |

### 3.2 Genetic algorithm

The GA parameters that have been set to obtain the optimal values of the PID gains are presented in table 3.2.

Table 3. 2 GA optimization parameter setting for PID controller

| Parameter           | Value        |
|---------------------|--------------|
| Population size     | 60           |
| Maximum generation  | 90           |
| Encoding            | Binary       |
| Selection           | Uniform      |
| Cross over          | Single point |
| Mutation            | Uniform      |
| Number of parameter | 3            |
| Lower bound         | [0 0 0]      |
| Upper bound         | [10 1 52]    |

### 3.3 Performance indices

This study has taken integral square error (ISE), integral absolute error (IAE), and integral time absolute error (ITAE) as an objective function of the genetic algorithm to tune the gains of the

PID controller. These objectives are also used to evaluate the overall performance of a control system. Equations for these performance indices are represented below [17].

$$\begin{cases} IAE = \int_0^{\infty} |e(t)| dt \\ ITAE = \int_0^{\infty} t |e(t)| dt \\ ISE = \int_0^{\infty} e^2(t) dt \end{cases} \quad (3.1)$$

## 4. Result and Discussion

### 4.1 Simulink models for the simulation

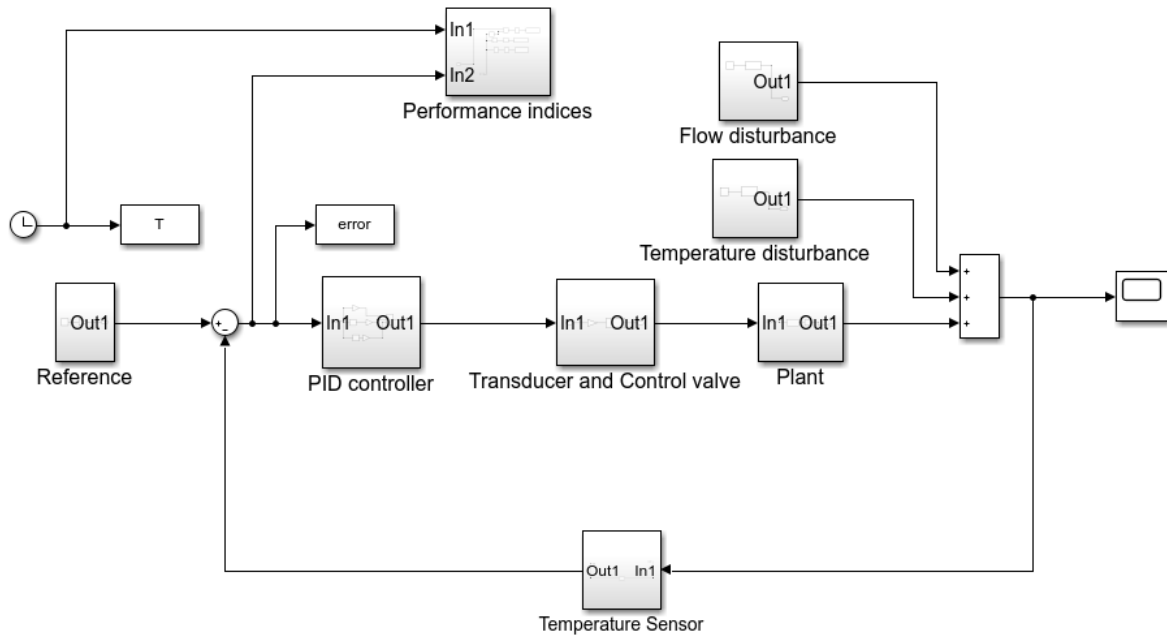


Figure 4. 1 Figure Simulink model of the proposed system

## 4.2 Open-loop response of the proposed system

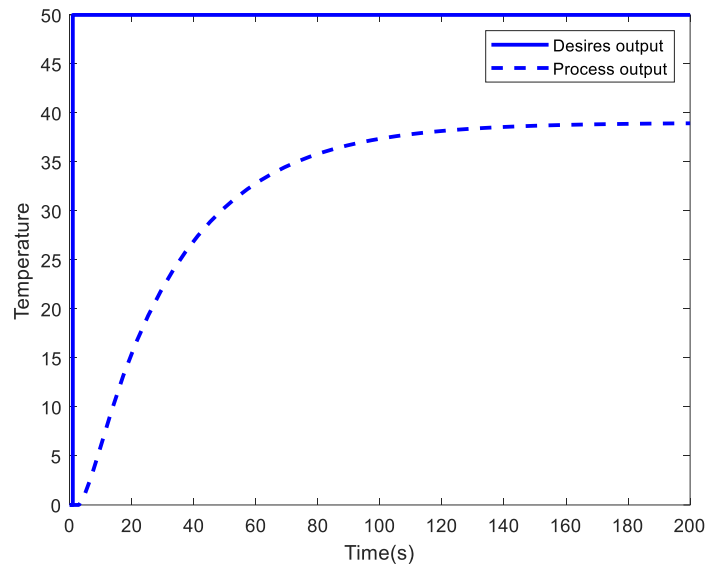


Figure 4. 2 Open-loop response of the proposed system

As can be observed in figure 4.2, the system doesn't settle around the set point, which is the main problem with the heat exchanger temperature control system. The above figure shows that the HE control system is highly subjected to the set point deviation issue and the amount of steady-state error is very large. Moreover, the system has taken a long time to settle, 120.0541sec. It is obvious that there is a need to implement a closed-loop control strategy.

## 4.3 Response of GA tuned PID under no disturbance condition

In table 4.1, GA tuned PID gains based on minimizing IAE, ITAE, and ISE objective functions are represented.

Table 4. 1 GA optimized PID gains for the three objective functions

| <div>J \ Gains</div> | K <sub>p</sub> | K <sub>i</sub> | K <sub>d</sub> |
|----------------------|----------------|----------------|----------------|
| J1=IAE               | 5.9982         | 0.1061         | 37.0577        |
| J2=ISE               | 5.9981         | 0.1239         | 49.9871        |
| J3=ITAE              | 4.2458         | 0.0813         | 28.1850        |

Based on the above parameter settings, the simulation result is shown in the figure below.

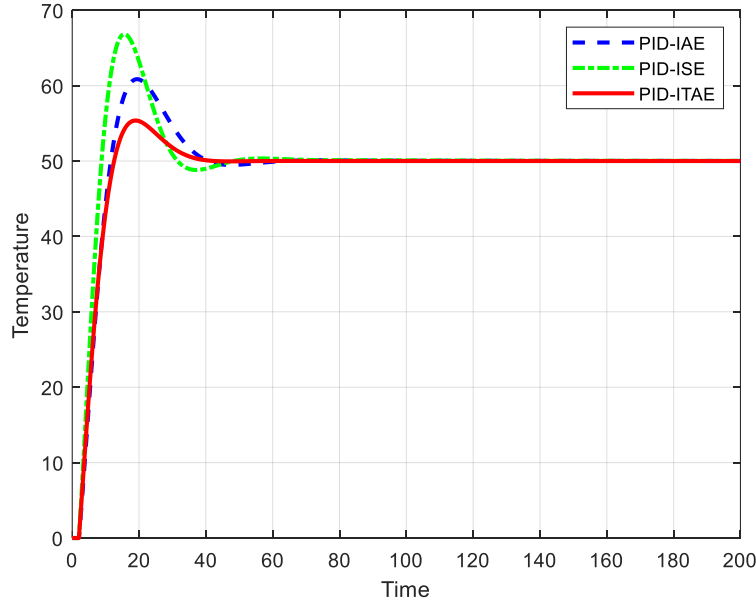


Figure 4. 3 GA tuned PID response under no disturbance condition

From the simulation result shown in figure 4.3, it can be understood that the PID controller tuned using a GA has eliminated the steady-state error. The performance of the PID controller tuned based on minimizing the ITAE objective function results in better performance than the one tuned based on minimizing ISE and the IAE objective function. The performance measures are summarized in table 4.2

Table 4. 2 Performance measure of GA\_PID controller under no disturbance condition

| Controllers | Performance measure |                      |       |
|-------------|---------------------|----------------------|-------|
|             | Settling time       | Percentage overshoot | IAE   |
| PID-ISE     | 40.3091sec          | 33.6136%             | 94.3  |
| PID-IAE     | 36.8457sec          | 21.6960%             | 108.7 |
| PID-ITAE    | 32.9412sec          | 10.7433%             | 111.2 |

From table 4.2, it can be clearly observed that GA PID tuned based on minimizing ISE objective function exhibits an overshoot of 33.61%. This overshoot has been reduced to 21.69%, an improvement of 35.5% by tuning the gains of PID based on minimizing the IAE objective function. Even though the overshoot has decreased to some extent, it can be further minimized to 10.7% by tuning the gains of the PID based on minimizing the ITAE objective function. In GA\_PID tuned

based on minimizing the ISE objective function, the settling time was 40.3091sec, whereas in GA-based PID tuned using the IAE objective function, the settling time has been minimized to 36.8457sec, an improvement of 8.6%. This overshoot is further minimized to 32.94sec using the ITAE objective function.

#### 4.4 Response of GA based PID subjected to flow variation disturbance

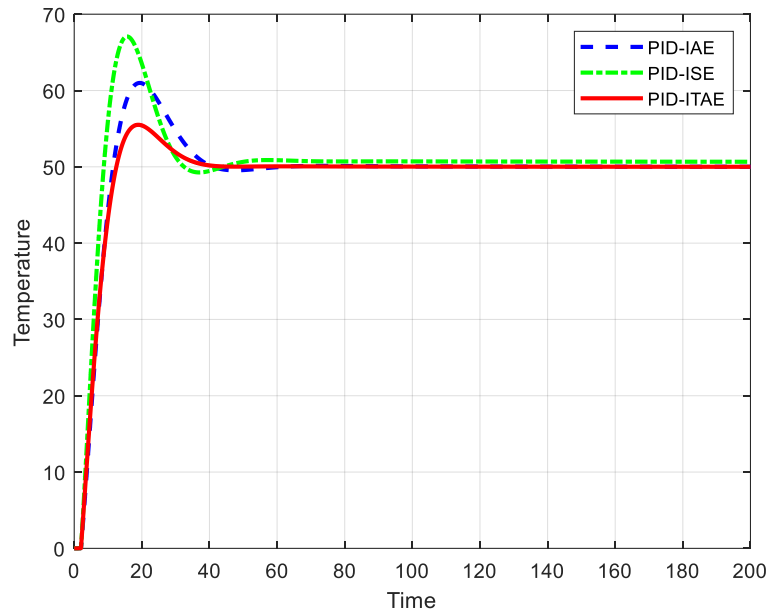


Figure 4. 4 GA tuned PID response under flow disturbance condition

From figure 4.4, it can be clearly observed that even with the existence of a flow disturbance effect, the performance of GA\_PID tuned based on minimizing ITAE objective function is superior to the others.

Table 4. 3 Performance measure of GA\_PID controller under flow disturbance condition

| Controllers | Performance measure |                      |       |
|-------------|---------------------|----------------------|-------|
|             | Settling time       | Percentage overshoot | IAE   |
| PID-ISE     | 41.5811sec          | 32.3967%             | 94.3  |
| PID-IAE     | 37.0463sec          | 21.9630%             | 109.3 |
| PID-ITAE    | 33.3557sec          | 11.0030%             | 111.9 |

From table 4.3, it can be observed that when the system is subjected to flow disturbance, the least settling time and overshoot are observed when the system is tuned based on minimizing the ITAE objective function, having numerical values of 33.3557sec and 11.003%, respectively. In contrast, the highest settling time and maximum overshoot are observed under the ISE objective function. The transient state performance of the IAE objective function is between ITAE and ISE. The other conclusion to be drawn from the above table is that the minimal IAE is observed when the ISE objective function is used.

#### 4.5 Response of GA tuned PID subjected to temperature variation disturbance

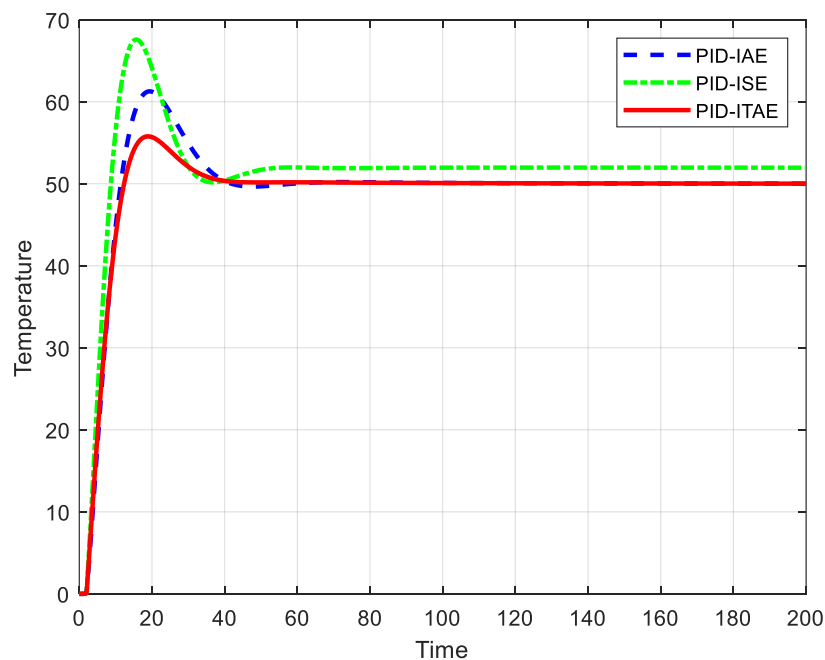


Figure 4. 5 GA tuned PID response under temperature disturbance condition

Figure 4.5 depicts that when the system is subjected to a temperature variation disturbance effect, the resultant steady state error is greater than the one that has occurred in the flow disturbance condition. The performance measures of the three objective functions are summarized in the table 4.4.

Table 4. 4 Performance measure of GA\_PID under temperature disturbance condition

| Controllers | Performance measure |                      |       |
|-------------|---------------------|----------------------|-------|
|             | Settling Time       | Percentage overshoot | IAE   |
| PID-ISE     | 43.6267sec          | 30.0560%             | 94.3  |
| PID-IAE     | 37.4588sec          | 22.4973%             | 110.6 |
| PID-ITAE    | 34.2404sec          | 11.5226%             | 112.1 |

From the above table, it can be observed that when the system is subjected to temperature variation disturbance, the least settling time and overshoot are observed under the ITAE objective function, having numerical values of 34.2404sec and 11.523%, respectively. In contrast, the highest settling time and maximum overshoot are observed under the ISE objective function, with numerical values of 43.6267sec and 30.0560%. The transient state performance of the IAE objective function is between ITAE and ISE. A minimal IAE is observed when ISE is used as an objective function.

#### 4.6 Response of GA tuned PID controller subjected to both disturbances

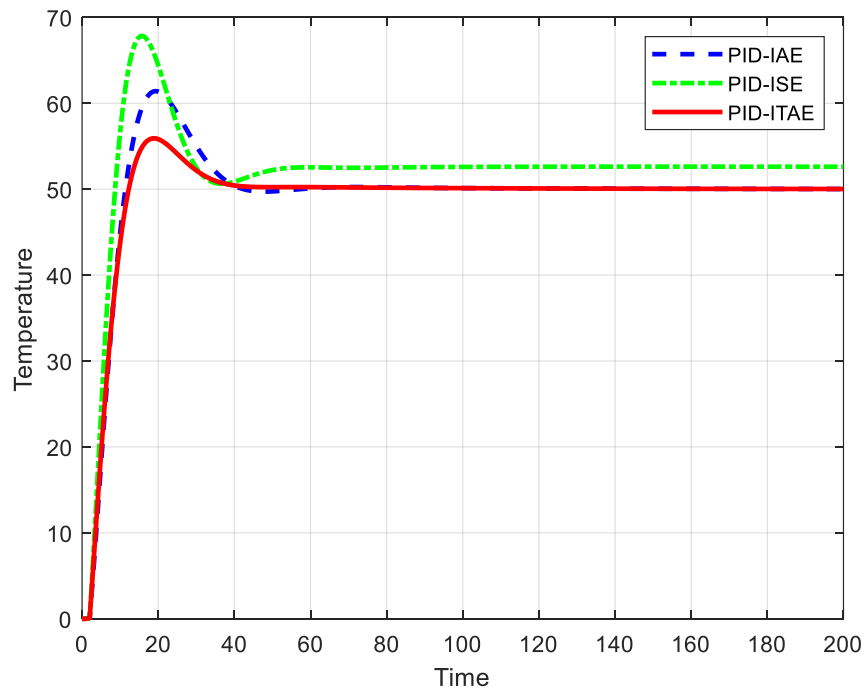


Figure 4. 6 GA tuned PID response under both disturbance condition

Table 4. 5 Performance measure of GA\_PID under both disturbance condition

| Controllers | Performance measure |                      |       |
|-------------|---------------------|----------------------|-------|
|             | Settling Time       | Percentage overshoot | IAE   |
| PID-ISE     | 44.5518sec          | 28.9300%             | 94.3  |
| PID-IAE     | 37.6711sec          | 22.7646%             | 111.2 |
| PID-ITAE    | 34.7165sec          | 11.7824%             | 112.7 |

From table 4.5 it can be clearly observed that GA PID tuned based on minimizing ISE objective function exhibits an overshoot of 28.93%. This overshoot has been reduced to 22.7646% by tuning the gains of PID based on minimizing the IAE objective function. Even though the overshoot has decreased to some extent, it can be further minimized to 11.7824% by tuning the gains of the PID based on minimizing the ITAE objective function. In GA\_PID tuned based on minimizing the ISE objective function, the settling time was 44.5518sec, whereas in GA\_PID tuned using the IAE objective function, the settling time has been minimized to 37.671sec. The settling time is further minimized to 34.7165sec using the ITAE objective function.

## 5. Conclusion and Recommendation

### 5.1 conclusion

This study has taken the STHE system to compare the performance of a GA PID controller tuned using different objective functions. The performance of the proposed controller has been compared under four scenarios, namely no disturbance, flow variation disturbance, temperature variation disturbance, and both disturbance conditions. The three objective functions that have been used for the genetic algorithm are IAE, ITAE, and ISE. Among the three objective functions, ITAE has provided the best performance with and without the presence of the disturbance effect in terms of transient response criteria, whereas the ISE objective function has provided the best performance in terms of error-based criteria when compared to the other two objective functions. The overall

performance of GA\_PID tuned based on minimizing ITAE objective function is good; however, it results in the presence of overshoot.

## 5.2 Recommendation

The parameters of the system specification have been taken from closely related work to compare the performance of the proposed controllers. Testing the controller's performance using real-time data taken from the industry is recommended. Also, the same work can be extended to other non-linear plants in process industries such as distillation columns, continuous stirred tank reactors, boilers, and electric furnaces.

The optimization method that has been used in this thesis is GA optimization techniques, and the work can be extended by implementing other optimization techniques such as Particle Swarm Optimization, the Grey Wolf optimization method, the Local Search Optimization Ant-Colony Optimization (ACO).

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